4d $\mathcal{N} = 2$ Theories from CY3 Compactification, Hitchin Systems, and Quantum Vertex Chiral Groups

Physics note for Volume III

Raeez Lorgat

April 2026

Contents

This note develops the physics of 4d $\mathcal{N} = 2$ theories from CY3 compactification through four layers of increasing structural depth, and connects the resulting picture to the quantum vertex chiral group framework of Volume III. The four layers are: (1) Type IIB on a CY3 X and the structure of the Coulomb branch; (2) the Hitchin system for local CY3 geometries and Seiberg–Witten theory; (3) the Kontsevich–Soibelman wall-crossing formula and its interpretation as gauge equivalence of Maurer–Cartan elements; (4) the Nekrasov partition function, its NS limit, quantum curves, and the character of the quantum vertex chiral group $G(X)$.

1 Type IIB on CY3: the 4d $\mathcal{N} = 2$ theory

1.1 The compactification

Let X be a compact Calabi–Yau threefold, i.e. a compact Kähler manifold of complex dimension 3 with $c_1(X) = 0$ and $h^{1,0}(X) = h^{2,0}(X) = 0$. The holonomy is $\mathrm{SU}(3) \subset \mathrm{SO}(6)$, and X carries a unique (up to scale) holomorphic $(3,0)$-form $\Omega \in H^0(X, \Omega_X^3)$ and a Ricci-flat Kähler metric in each Kähler class.

Type IIB string theory compactified on X yields a 4d $\mathcal{N} = 2$ supergravity theory. The massless spectrum in four dimensions is determined by the Hodge numbers of X:

- The gravity multiplet (always present);
- $h^{2,1}(X)$ vector multiplets, parametrized by complex structure deformations of X;
- $h^{1,1}(X)$ hypermultiplets, parametrized by complexified Kähler deformations of X;
- One universal hypermultiplet (the dilaton-axion).

The $\mathcal{N} = 2$ moduli space factorizes (at tree level) as

$$\mathcal{M} = \mathcal{M}_V \times \mathcal{M}_H, \tag{1}$$

where $\mathcal{M}_V$ is a special Kähler manifold of complex dimension $h^{2,1}(X)$ (the vector multiplet moduli space) and $\mathcal{M}_H$ is a quaternionic Kähler manifold of real dimension $4(h^{1,1}(X)+1)$ (the hypermultiplet moduli space). This factorization is protected by $\mathcal{N}=2$ supersymmetry: the prepotential $\mathcal{F}$ governing $\mathcal{M}_V$ receives no perturbative corrections from hypermultiplet scalars, and vice versa.

1.2 The Coulomb branch as complex structure moduli

The vector multiplet moduli space $\mathcal{M}_V$ is identified with the Coulomb branch of the 4d theory. For Type IIB on X, this is the complex structure moduli space:

$$\mathcal{M}_{\text{Coulomb}} = \mathcal{M}_{\text{cx}}(X). \tag{2}$$

A point $u \in \mathcal{M}_{\text{Coulomb}}$ specifies a complex structure on X, hence a specific $\Omega_u \in H^{3,0}(X_u)$. The special Kähler geometry on $\mathcal{M}_V$ is determined by the *periods* of Ω_u: choose a symplectic basis $\{A^I, B_J\}$ of $H_3(X,\mathbb{Z})$ (with $A^I \cdot B_J = \delta^I_J$, $I, J = 0, \ldots, h^{2,1}$), and define

$$z^I(u) = \oint_{A^I} \Omega_u, \qquad \mathcal{F}_J(u) = \oint_{B_J} \Omega_u. \tag{3}$$

The z^I serve as special coordinates on $\mathcal{M}_V$, and $\mathcal{F}_J = \partial\mathcal{F}/\partial z^J$ for a holomorphic prepotential $\mathcal{F}(z)$ (homogeneous of degree 2 in the z^I).

Remark 1.1 (Mirror symmetry interchange). Under mirror symmetry $X \leftrightarrow X^\vee$, the roles of complex structure and Kähler moduli are exchanged. For Type IIA on $X^\vee$, the Coulomb branch is $\mathcal{M}_{\text{Käh}}(X^\vee)$, and the prepotential is computed by genus-0 Gromov–Witten invariants. The physical equivalence IIB/$X \cong$ IIA/$X^\vee$ is the origin of mirror symmetry at the level of effective 4d theories.

1.3 The BPS spectrum: D3-branes on special Lagrangians

The BPS states of the 4d $\mathcal{N}=2$ theory arise from D3-branes wrapping special Lagrangian 3-cycles in X. A special Lagrangian $L \subset X$ is a 3-dimensional submanifold satisfying:

(i) $\omega|_L = 0$ (Lagrangian condition: L is isotropic for the Kähler form);

(ii) $\text{Im}(e^{-i\theta}\Omega)|_L = 0$ for some phase θ (special condition: calibrated by the real part of a rotated Ω).

The mass of the BPS state associated to a class $\gamma \in H_3(X,\mathbb{Z})$ is

$$M_\gamma(u) = |Z_\gamma(u)| = \left| \oint_\gamma \Omega_u \right|, \tag{4}$$

where $Z_\gamma(u)$ is the central charge of the $\mathcal{N}=2$ algebra. This is a function on the Coulomb branch: as u varies, the central charges rotate, and BPS bound states can form or decay.

The BPS index (second helicity supertrace) is

$$\Omega(\gamma; u) = \text{Tr}_{\mathcal{A}_\gamma}(-1)^{2J_3}(2J_3)^2, \tag{5}$$

where $\mathcal{A}_\gamma$ is the Hilbert space of one-particle states of charge γ. This index is piecewise constant in u but jumps across real-codimension-1 walls in $\mathcal{M}_{\text{Coulomb}}$ — the *walls of marginal stability.*

Remark 1.2 (Charge lattice). The charge lattice $\Gamma = H_3(X,\mathbb{Z})$ carries the symplectic intersection form $\langle\gamma_1,\gamma_2\rangle = \gamma_1\cdot\gamma_2$. In the Volume III language, Γ is the lattice $\Lambda(X)$ of the generalized root datum $\mathcal{R}(X)$, and the BPS indices $\Omega(\gamma;u)$ are the root multiplicities $\mathrm{mult}(\gamma)$.

1.4 The derived category perspective

From the categorical perspective, the BPS states are objects of the derived category $D^b(\mathrm{Coh}(X))$ (B-branes) or equivalently the Fukaya category $\mathrm{Fuk}(X)$ (A-branes, via HMS). The stability condition σ_u on $D^b(\mathrm{Coh}(X))$ corresponding to a point $u\in\mathcal{M}_{\mathrm{Coulomb}}$ determines which objects are σ_u-stable — these are the BPS states. The central charge $Z_\gamma(u) = \oint_\gamma \Omega_u$ is exactly the central charge function of the Bridgeland stability condition.

In the quantum vertex chiral group framework: the CY-to-chiral functor $\Phi\colon \mathrm{CY}_3\text{-Cat}\to E_2\text{-ChirAlg}$ applied to $D^b(\mathrm{Coh}(X))$ or $\mathrm{Fuk}(X)$ produces the quantum chiral algebra $A_X = \Phi(\mathcal{C}_X)$. The BPS spectrum is encoded in the root multiplicities of $G(X)$.

2 The Hitchin system and Seiberg–Witten theory

2.1 Local CY3 geometries and gauge theory engineering

For local (noncompact) CY3 geometries, one engineers pure $\mathcal{N}=2$ gauge theories in four dimensions. The prototypical construction: let C be a compact Riemann surface and G a reductive group. The local CY3 is the total space of the cotangent bundle $T^*\Sigma$ (or more generally, a conic bundle over C determined by a Hitchin spectral curve).

Example 2.1 (Pure $\mathrm{SU}(N)$ gauge theory)**.** Consider the local CY3 geometry

$$X_{G,C} = T^*C\times\mathbb{C}\quad\text{(fibered appropriately)},\tag{6}$$

where C is a curve of appropriate genus. The compactification of Type IIB on $X_{G,C}$ engineers a 4d $\mathcal{N}=2$ gauge theory with gauge group G and coupling constant determined by the complex structure of C.

More precisely, the local CY3 is constructed via a system of D5-branes wrapping C (in the IIB picture) or M5-branes wrapping C (in the M-theory picture). The 6d $(2,0)$ theory of type $\mathfrak{g} = \mathrm{Lie}(G)$ on $\mathbb{R}^{3,1}\times C$ produces, upon compactification on C, the 4d $\mathcal{N}=2$ theory of class $\mathcal{S}[G,C,\mathcal{D}]$, where $\mathcal{D}$ denotes the defect data (punctures on C, with prescribed boundary conditions labeled by nilpotent orbits in $\mathfrak{g}$).

2.2 The Hitchin system

The Coulomb branch of the class $\mathcal{S}$ theory is governed by the Hitchin system. Let C be a smooth projective curve of genus g_C, and let G be a complex reductive group with Lie algebra $\mathfrak{g}$.

Definition 2.2 (Hitchin moduli space)**.** The *Hitchin moduli space* $\mathcal{M}_H(C,G)$ is the moduli space of solutions to Hitchin's equations on C: pairs (E,φ) where E is a principal G-bundle on C and $\varphi\in H^0(C,\mathrm{ad}(E)\otimes K_C)$ is a Higgs field (a holomorphic section of the adjoint bundle twisted by the canonical bundle). Equivalently, $\mathcal{M}_H(C,G)$ parametrizes S-equivalence classes of semistable G-Higgs bundles on C.

The Hitchin moduli space is a hyperkähler manifold of complex dimension $2\dim G \cdot (g_C - 1)$ (for $g_C \geq 2$). It admits three complex structures I, J, K:

- In complex structure I: $\mathcal{M}_H$ parametrizes Higgs bundles (E,φ);
- In complex structure J: $\mathcal{M}_H$ parametrizes flat $G_{\mathbb{C}}$-connections (the de Rham moduli space $\mathcal{M}_{\mathrm{dR}}(C,G)$), via the nonabelian Hodge correspondence $(E,\varphi) \mapsto \nabla = D_E + \varphi + \varphi^*$;
- In complex structure K: $\mathcal{M}_H$ parametrizes flat connections with a different reality condition.

Definition 2.3 (Hitchin fibration)**.** The *Hitchin fibration* is the map

$$\pi\colon \mathcal{M}_H(C,G) \longrightarrow \mathcal{A}_H = \bigoplus_{i=1}^{\mathrm{rk}(G)} H^0(C, K_C^{\otimes d_i}), \tag{7}$$

where $d_1, \dots, d_{\mathrm{rk}(G)}$ are the degrees of the fundamental G-invariant polynomials on $\mathfrak{g}$. The map sends $(E,\varphi) \mapsto (\mathrm{Tr}(\varphi^{d_1}), \dots, \mathrm{Tr}(\varphi^{d_{\mathrm{rk}(G)}}))$. The base $\mathcal{A}_H$ is called the *Hitchin base*.

The Hitchin fibration is a completely integrable system: the generic fiber $\pi^{-1}(a)$ is an abelian variety (a Prym variety of the spectral curve Σ_a), the dimension of $\mathcal{A}_H$ equals half the dimension of $\mathcal{M}_H$, and the Hitchin Hamiltonians Poisson-commute.

2.3 The spectral curve as Seiberg–Witten curve

The fundamental identification linking the Hitchin system to 4d $\mathcal{N} = 2$ gauge theory:

Theorem 2.4 (Donagi–Witten, Martinec–Warner, Gorsky–Krichever–Marshakov–Mironov–Morozov)**.** *The Seiberg–Witten curve Σ_{SW} of the 4d $\mathcal{N} = 2$ gauge theory of class $\mathcal{S}[G,C]$ is the spectral curve of the Hitchin system on (C,G). Specifically:*

(i) The SW curve is

$$\Sigma_a = \Big\{(x,p) \in T^*C \;\Big|\; \det(p \cdot \mathrm{id} - \varphi(x)) = 0\Big\} \subset T^*C, \tag{8}$$

a branched cover $\Sigma_a \to C$ of degree $\mathrm{rk}(G)$, determined by the point $a \in \mathcal{A}_H$ (the Hitchin base coordinates = the Coulomb branch parameters);

(ii) The SW differential is $\lambda_{\mathrm{SW}} = p\,dx$, the tautological 1-form on T^*C restricted to Σ_a;

(iii) The periods of λ_{SW} over the cycles of Σ_a give the special coordinates and their duals:

$$a_I = \oint_{\alpha_I} \lambda_{\mathrm{SW}}, \qquad a_{D,J} = \oint_{\beta_J} \lambda_{\mathrm{SW}} = \frac{\partial \mathcal{F}}{\partial a_J}; \tag{9}$$

(iv) The Hitchin base $\mathcal{A}_H$ is the Coulomb branch $\mathcal{M}_{\mathrm{Coulomb}}$, and the Hitchin fibration π parametrizes the family of SW curves.

Example 2.5 (Pure SU(2))**.** Take $G = \mathrm{SU}(2)$, $C = \mathbb{P}^1$. The Hitchin base is $\mathcal{A}_H = H^0(\mathbb{P}^1, K^{\otimes 2}) \cong \mathbb{C}$ (one Coulomb branch parameter u). The spectral curve in $T^*\mathbb{P}^1$ is

$$\Sigma_u\colon \quad p^2 = u + \Lambda^2 x + \Lambda^{-2} x^{-1} \tag{10}$$

(in appropriate coordinates), recovering the Seiberg–Witten curve. The discriminant locus $\{u : \Sigma_u \text{ is singular}\}$ gives the positions of the monopole and dyon singularities in the Coulomb branch, where BPS states become massless.

Example 2.6 (SU(N) with fundamental matter)**.** For $G = \mathrm{SU}(N)$ with N_f fundamental hypermultiplets, the curve $C = \mathbb{P}^1$ carries N_f punctures. The spectral curve is an N-fold cover of $\mathbb{P}^1$:

$$\det(p \cdot \mathrm{id}_N - \varphi(x)) = p^N + u_2(x) p^{N-2} + \cdots + u_N(x) = 0, \tag{11}$$

where the $u_k(x)$ are meromorphic k-differentials on $\mathbb{P}^1$ with poles at the punctures (of orders determined by the matter content).

2.4 The Hitchin system in the CY framework

The Hitchin moduli space $\mathcal{M}_H(C, G)$ is itself a (noncompact) hyperkähler manifold, hence in particular a (noncompact) holomorphic symplectic manifold. The total space of the spectral cover $\Sigma_a \to C$ inside T^*C is a local CY: T^*C has trivial canonical bundle (the holomorphic symplectic form $\omega = dp \wedge dx$ trivializes it).

In the framework of Volume III, the Hitchin system provides the following data for the quantum vertex chiral group:

- The lattice $\Lambda(X)$ is the homology $H_1(\Sigma_a, \mathbb{Z})$ of the spectral curve (the charge lattice of the 4d theory);
- The Hitchin base $\mathcal{A}_H$ parametrizes the complex structure deformations (the Coulomb branch);
- The Hitchin fibers are the abelian varieties on which the BPS states live;
- The BPS indices $\Omega(\gamma; a)$ determine the root multiplicities of $G(X)$.

3 Wall-crossing and the Kontsevich–Soibelman formula

3.1 Walls of marginal stability

Fix a charge lattice Γ with antisymmetric pairing $\langle -, - \rangle$ and a central charge function $Z\colon \Gamma \to \mathbb{C}$ (varying holomorphically over $\mathcal{M}_{\mathrm{Coulomb}}$). A *wall of marginal stability* for charges $\gamma_1, \gamma_2 \in \Gamma$ is the locus

$$\mathcal{W}(\gamma_1, \gamma_2) = \left\{ u \in \mathcal{M}_{\mathrm{Coulomb}} \;\middle|\; \frac{Z_{\gamma_1}(u)}{Z_{\gamma_2}(u)} \in \mathbb{R}_{>0} \right\}, \tag{12}$$

i.e. the set of points where Z_{γ_1} and Z_{γ_2} are aligned in the complex plane. Across such a wall, BPS bound states of charge $\gamma = \gamma_1 + \gamma_2$ can form or decay, and the BPS index $\Omega(\gamma; u)$ jumps.

3.2 The Kontsevich–Soibelman wall-crossing formula

Kontsevich and Soibelman formulated a universal wall-crossing formula governing the jumps.

Definition 3.1 (BPS automorphisms)**.** For each $\gamma \in \Gamma$ with $\Omega(\gamma) \neq 0$, define the automorphism $\mathcal{K}_\gamma$ of the quantum torus algebra $\mathbb{C}[\Gamma]$ (with generators X_γ satisfying $X_{\gamma_1} X_{\gamma_2} = (-1)^{\langle \gamma_1, \gamma_2 \rangle} X_{\gamma_1+\gamma_2}$) by:

$$\mathcal{K}_\gamma = \prod_{n \geq 1} \left(1 - (-1)^n X_{n\gamma}\right)^{n\,\Omega(n\gamma)} . \tag{13}$$

For the primitive case $\Omega(\gamma) = 1$ (a single hypermultiplet), this reduces to the Kontsevich–Soibelman symplectomorphism $\mathcal{K}_\gamma(X_{\gamma'}) = X_{\gamma'}(1 + X_\gamma)^{\langle \gamma, \gamma' \rangle}$.

Theorem 3.2 (Kontsevich–Soibelman)**.** *The ordered product*

$$\mathcal{A}(u) = \overset{\curvearrowleft}{\prod_{\gamma : \arg Z_\gamma(u) \downarrow}} \mathcal{K}_\gamma^{\Omega(\gamma; u)} \tag{14}$$

taken over all BPS charges γ in decreasing order of the phase $\arg Z_\gamma(u)$*, is* invariant *under continuous deformation of u across walls of marginal stability. That is, although the individual $\Omega(\gamma; u)$ jump and the ordering of phases changes, the total product $\mathcal{A}(u)$ remains constant.*

Remark 3.3 (Categorification)*.* The KS formula has a categorical interpretation via the theory of Bridgeland stability conditions on $D^b(\mathrm{Coh}(X))$ (or $\mathrm{Fuk}(X)$). As the stability condition varies, the set of stable objects changes, but the Stokes factors of the associated Riemann–Hilbert problem remain constant as a product. This is the Stokes phenomenon for the differential equation governing the "tt*-geometry" of the Coulomb branch.

3.3 Wall-crossing as gauge equivalence of MC elements

In the framework of Volumes I–III, the KS wall-crossing formula admits a natural interpretation as gauge equivalence of Maurer–Cartan elements.

Construction 3.4 (MC interpretation of wall-crossing)**.** The key identifications:

(i) The L_∞-algebra controlling the deformation problem is the *scattering diagram L_∞-algebra* $\mathfrak{L}_\Gamma$, built from the charge lattice Γ with its antisymmetric form. Its Maurer–Cartan equation governs consistent scattering diagrams on $\mathbb{R}^2 \cong \mathbb{C}$ (the Z-plane).

(ii) A BPS spectrum $\{\Omega(\gamma; u)\}_{\gamma \in \Gamma}$ at a point $u \in \mathcal{M}_{\mathrm{Coulomb}}$ determines a Maurer–Cartan element

$$\Theta_u = \sum_{\gamma \in \Gamma} \Omega(\gamma; u) \cdot e_\gamma \in \mathfrak{L}_\Gamma, \tag{15}$$

where e_γ are the generators associated to charges γ.

(iii) Crossing a wall $\mathcal{W}(\gamma_1, \gamma_2)$ corresponds to a *gauge transformation* $\Theta_u \mapsto \Theta_{u'} = g \cdot \Theta_u$ in the L_∞-algebra, where g is determined by the KS symplectomorphism $\mathcal{K}_\gamma$.

(iv) The KS invariant $\mathcal{A}(u)$ is the *gauge equivalence class* $[\Theta_u] \in \mathrm{MC}(\mathfrak{L}_\Gamma)/\mathrm{gauge}$.

This is the precise connection to the universal MC element Θ_A of Volume I. The quantum vertex chiral group $G(X)$ has a MC element Θ_{A_X} encoding the BPS spectrum, and variation of the Coulomb branch parameter u (= the complex structure of X) produces gauge-equivalent MC elements. The shadow obstruction tower

$$\Theta_{A_X}^{\leq r} = \sum_{\substack{\gamma\in\Gamma \\ \text{arity}\leq r}} \Omega(\gamma)\cdot e_\gamma \tag{16}$$

at successive arities recovers the BPS spectrum at successive levels of complexity: arity 2 gives the leading central charge (the "Weyl vector" contribution κ), arity 3 gives the first bound-state corrections, and the full tower gives the complete BPS generating function — the automorphic form.

Proposition 3.5 (Wall-crossing = shadow obstruction tower gauge equivalence)**.** *The KS wall-crossing formula for the 4d $\mathcal{N}=2$ theory from CY3 X is equivalent to the statement that the MC elements Θ_u, as u varies over $\mathcal{M}_{\mathrm{Coulomb}} = \mathcal{M}_{\mathrm{cx}}(X)$, are gauge-equivalent in $\mathfrak{L}_{\Gamma(X)}$. In particular:*

- *(i) The gauge equivalence class $[\Theta_u]$ is a topological invariant of the 4d theory;*
- *(ii) The individual $\Omega(\gamma;u)$ are the "coordinates" of Θ_u in a particular gauge (= a choice of "chamber" in the Coulomb branch);*
- *(iii) Wall-crossing is a change of gauge; the KS product $\mathcal{A}(u)$ is the gauge-invariant content.*

Remark 3.6 (Scattering diagrams and tropical geometry)*.* The scattering diagram technology of Gross–Siebert and Kontsevich–Soibelman provides the combinatorial framework for encoding the MC element. A scattering diagram $\mathfrak{D}$ on $\mathbb{R}^2$ consists of rays (walls) emanating from the origin, decorated with automorphisms of the quantum torus. The consistency condition (trivial monodromy around the origin) is exactly the MC equation for Θ. This has been made precise by Bridgeland in terms of Stokes data and by Gaiotto–Moore–Neitzke in terms of the "spectral network" formalism.

3.4 The motivic DT invariants and the Hall algebra

The BPS indices $\Omega(\gamma;u)$ are the numerical shadows of a richer structure: the *motivic Donaldson–Thomas invariants*. Kontsevich and Soibelman introduced motivic DT invariants $\Omega^{\mathrm{mot}}(\gamma;u) \in K_0(\mathrm{Var}/\mathbb{C})[\mathbb{L}^{-1/2}]$ (classes in the motivic ring) lifting the numerical $\Omega(\gamma;u)$.

The motivic Hall algebra $H(\mathcal{C})$ of a CY3 category $\mathcal{C}$ (in the sense of Joyce, Kontsevich–Soibelman, or Bridgeland) provides the natural home for the motivic wall-crossing formula. The multiplication in $H(\mathcal{C})$ encodes extensions of objects, and the integration map (motivic measure) $H(\mathcal{C}) \to$ quantumtorus descends to the KS symplectomorphisms.

In the Volume III language: the motivic Hall algebra is the chain-level refinement of the quantum vertex chiral group. The critical CoHA ("Toric CY3 and Critical CoHAs" chapter of Volume III) is the cohomological shadow of this motivic structure, and the passage

$$H^{\mathrm{mot}}(\mathcal{C}) \xrightarrow{\text{cohomological}} \mathcal{H}(\mathcal{C}) = \mathrm{CoHA} \xrightarrow{\text{positive half}} G^+(X) \tag{17}$$

recovers the E_1-sector (positive half) of $G(X)$.

4 The Nekrasov partition function and quantum curves

4.1 The Ω-background

Nekrasov's partition function computes the exact prepotential of the 4d $\mathcal{N}=2$ theory by introducing the Ω-*background*: a deformation of $\mathbb{R}^4$ that localizes the path integral to fixed points of a torus action.

Let $T = U(1)_{\epsilon_1} \times U(1)_{\epsilon_2}$ act on $\mathbb{R}^4 \cong \mathbb{C}^2$ by $(z_1, z_2) \mapsto (e^{i\epsilon_1} z_1, e^{i\epsilon_2} z_2)$. The *Nekrasov partition function* for a 4d $\mathcal{N}=2$ gauge theory with gauge group G and Coulomb branch parameters $\vec{a} = (a_1, \ldots, a_{\mathrm{rk}(G)})$ is

$$Z_{\mathrm{Nek}}(\epsilon_1, \epsilon_2, \vec{a}; \mathfrak{q}) = \sum_{\vec{\lambda}} \mathfrak{q}^{|\vec{\lambda}|} \, z_{\vec{\lambda}}(\epsilon_1, \epsilon_2, \vec{a}), \tag{18}$$

where the sum is over tuples $\vec{\lambda} = (\lambda^{(1)}, \ldots, \lambda^{(\mathrm{rk}(G))})$ of Young diagrams (= fixed points of the T-action on the instanton moduli space $\mathcal{M}_{\mathrm{inst}}(G,k) = \mathcal{M}(k, \mathrm{rk}(G))$, parametrized by $\mathrm{rk}(G)$-tuples of partitions of total size k), $\mathfrak{q} = e^{2\pi i \tau}$ is the instanton counting parameter, and $z_{\vec{\lambda}}$ are the equivariant weights computed by the Atiyah–Bott localization formula.

Theorem 4.1 (Nekrasov)**.** *The Seiberg–Witten prepotential is recovered in the classical limit:*

$$\mathcal{F}(\vec{a}) = \lim_{\epsilon_1, \epsilon_2 \to 0} \epsilon_1 \epsilon_2 \log Z_{\mathrm{Nek}}(\epsilon_1, \epsilon_2, \vec{a}; \mathfrak{q}). \tag{19}$$

4.2 Nekrasov partition function as refined CY3 partition function

The Nekrasov partition function is a *refinement* of the CY3 topological string partition function. For the local CY3 $X_{G,C}$ engineering the gauge theory:

(i) The topological string partition function $Z_{\mathrm{top}}(g_s, t)$ (where g_s is the string coupling and t are Kähler moduli) satisfies

$$Z_{\mathrm{top}}(g_s, t) = Z_{\mathrm{Nek}}(\epsilon_1 = g_s, \epsilon_2 = -g_s, \vec{a}(t)), \tag{20}$$

i.e. the unrefined ($\epsilon_1 = -\epsilon_2$) Nekrasov partition function equals the topological string partition function.

(ii) The *refined* topological string partition function (Iqbal–Kozcaz–Vafa, Hollowood–Iqbal–Vafa) is

$$Z^{\mathrm{ref}}_{\mathrm{top}}(\epsilon_1, \epsilon_2, t) = Z_{\mathrm{Nek}}(\epsilon_1, \epsilon_2, \vec{a}(t)), \tag{21}$$

the full two-parameter Nekrasov partition function. The refinement breaks the symmetry $\epsilon_1 \leftrightarrow \epsilon_2$ and corresponds to a "motivic" lift of the DT/GW invariants.

(iii) The genus expansion of the topological string is

$$\log Z_{\mathrm{top}} = \sum_{g \geq 0} g_s^{2g-2} \, \mathcal{F}_g(t), \tag{22}$$

where $\mathcal{F}_0$ is the prepotential, $\mathcal{F}_1$ is the genus-1 free energy (related to κ and the Ray–Singer torsion), and $\mathcal{F}_g$ for $g \geq 2$ are the higher-genus Gromov–Witten/DT invariants.

In the Volume I language: $\mathcal{F}_g$ is the genus-g obstruction; on the uniform-weight lane, $\mathrm{obs}_g = \kappa \cdot \lambda_g$ (for multi-weight algebras at $g \geq 2$, cross-channel corrections appear; see Vol I, AP32).

Remark 4.2 (Refined vs. unrefined in Volume III). The unrefined limit $\epsilon_1 = -\epsilon_2 = g_s$ corresponds to the topological string, where the genus expansion is controlled by a single parameter g_s ($= \hbar$ in the conventions of Volume I, up to the reconciliation $\hbar_{\mathrm{Feynman}} = \hbar^2_{\mathrm{here}}$). The full two-parameter refinement (ϵ_1, ϵ_2) is a finer invariant: it distinguishes the two $U(1)$ factors of the Cartan of $\mathrm{SO}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, and the "motivic" DT invariants of Kontsevich–Soibelman are the natural home for this data.

4.3 The NS limit and quantum curves

The Nekrasov–Shatashvili (NS) limit is the half-Ω-background:

$$\epsilon_2 \to 0, \qquad \epsilon_1 = \hbar \text{ fixed.} \tag{23}$$

In this limit, the partition function develops an essential singularity:

$$\log Z_{\mathrm{Nek}}(\hbar, \epsilon_2, \vec{a}) \xrightarrow{\epsilon_2 \to 0} \frac{1}{\epsilon_2} \mathcal{W}(\hbar, \vec{a}) + O(1), \tag{24}$$

where $\mathcal{W}(\hbar, \vec{a})$ is the *twisted effective superpotential* (or "NS free energy").

Theorem 4.3 (Nekrasov–Shatashvili)**.** *In the NS limit:*

(i) The saddle-point equation

$$\exp\left(\frac{\partial \mathcal{W}}{\partial a_I}\right) = 1 \tag{25}$$

determines the "quantum" Coulomb branch parameters $a_I(\hbar)$ as functions of $\hbar$;

(ii) The Seiberg–Witten curve Σ_{SW} is quantized*: the classical equation*

$$\det(p \cdot \mathrm{id} - \varphi(x)) = 0 \tag{26}$$

is promoted to a quantum curve *— a differential equation*

$$\widehat{\Sigma}_\hbar: \qquad \det\left(\hbar \frac{d}{dx} \cdot \mathrm{id} - \varphi(x)\right) \psi(x) = 0, \tag{27}$$

obtained by the canonical quantization $p \mapsto \hbar\, d/dx$ of the cotangent bundle T^*C;

(iii) The NS partition function $\psi(x) = Z^{\mathrm{NS}}_{\mathrm{Nek}}(\hbar, x)$ (with x a position variable on C) is a solution of the quantum curve equation (**??**).

Example 4.4 (Quantum SU(2) curve)**.** For pure SU(2), the classical SW curve is $p^2 = u + \Lambda^2 x + \Lambda^{-2} x^{-1}$. The quantum curve is the second-order ODE

$$\left[\hbar^2 \frac{d^2}{dx^2} - u - \Lambda^2 x - \Lambda^{-2} x^{-1}\right] \psi(x) = 0. \tag{28}$$

This is a confluent Heun equation (degenerate Mathieu equation), whose monodromy data encodes the exact spectrum.

Example 4.5 (Quantum SU(N) curve and $\mathcal{W}$-algebra)**.** For SU(N), the quantum spectral curve is an N-th order ODE

$$\left[\left(\hbar\frac{d}{dx}\right)^N + u_2(x)\left(\hbar\frac{d}{dx}\right)^{N-2} + \cdots + u_N(x)\right]\psi(x) = 0. \tag{29}$$

This is the *oper* equation: the quantization of the Hitchin spectral curve produces an ${}^LG$-oper on C (where ${}^LG$ is the Langlands dual group). The quantum curve is the null vector equation of the $\mathcal{W}_N$-algebra at level $k = -N + \hbar^{-1}$ (the critical level is $k = -h^\vee = -N$; the NS parameter $\hbar$ provides the shift from critical).

4.4 Connection to quantum vertex chiral groups

The Nekrasov partition function connects to the quantum vertex chiral group $G(X)$ through the following chain of identifications:

Construction 4.6 (Nekrasov partition function as character of $G(X)$)**.** Let X be a (local) CY3 engineering a 4d $\mathcal{N} = 2$ theory with gauge group G.

(i) The quantum vertex chiral group $G(X)$ acts on a Fock space $\mathcal{F}_{\vec{a}}$ parametrized by the Coulomb branch parameters $\vec{a}$.

(ii) The Nekrasov partition function is a character:

$$Z_{\text{Nek}}(\epsilon_1, \epsilon_2, \vec{a}; \mathfrak{q}) = \mathrm{Tr}_{\mathcal{F}_{\vec{a}}}\, \mathfrak{q}^{L_0}\, q_1^{J_1}\, q_2^{J_2}, \tag{30}$$

where $q_i = e^{i\epsilon_i}$ are the equivariant fugacities and J_1, J_2 are the generators of the $U(1)^2$ action. This is the graded character of the vacuum module of $G(X)$.

(iii) For toric CY3: $G(X)$ is the affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$, and the Nekrasov partition function is a character of the Yangian. For $X = \mathbb{C}^3$: $G(X) = Y(\widehat{\mathfrak{gl}}_1) \cong \mathcal{W}_{1+\infty}$, and Z_{Nek} is the MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$, the generating function for 3d partitions.

(iv) For the $K3 \times E$ tower: $G(X)$ is the BKM superalgebra $\mathfrak{g}_{\Delta_5}$, and the partition function involves the Igusa cusp form Δ_5. The Nekrasov partition function (= the refined topological vertex computation) should be a character of $\mathfrak{g}_{\Delta_5}$.

Conjecture 4.7 (Nekrasov partition function as character)**.** *For any CY3 X engineering a 4d $\mathcal{N} = 2$ theory, the Nekrasov partition function $Z_{\text{Nek}}(\epsilon_1, \epsilon_2, \vec{a}; \mathfrak{q})$ is the graded character of the vacuum module of the quantum vertex chiral group $G(X)$:*

$$Z_{\text{Nek}}(\epsilon_1, \epsilon_2, \vec{a}; \mathfrak{q}) = \chi_{G(X)}(\mathcal{F}_{\vec{a}}; q_1, q_2, \mathfrak{q}). \tag{31}$$

In the unrefined limit $\epsilon_1 = -\epsilon_2 = g_s$, this reduces to the topological string partition function, which is the denominator identity of $G(X)$.

Remark 4.8 (Evidence). The evidence for Conjecture **??**:

(1) For $X = \mathbb{C}^3$: the Nekrasov partition function is $M(q) = \prod(1-q^n)^{-n}$, which IS the character of the MacMahon module of $\mathcal{W}_{1+\infty} = Y(\widehat{\mathfrak{gl}}_1)$. This is proved (Schiffmann–Vasserot, Maulik–Okounkov).
(2) For the resolved conifold: Z_{Nek} factors as a product of MacMahon functions times a quantum dilogarithm, matching the character of $Y(\widehat{\mathfrak{gl}}_{1|1})$.
(3) For $K3 \times E$: the topological string partition function involves Δ_5^{-1} (the inverse Igusa cusp form), which is the character of the BKM superalgebra denominator formula inverted — i.e. the Fock space character.
(4) The AGT correspondence (Alday–Gaiotto–Tachikawa) identifies Z_{Nek} for SU(2) gauge theories with conformal blocks of the Virasoro algebra at $c = 1 + 6(\sqrt{\beta} - 1/\sqrt{\beta})^2$ where $\beta = \epsilon_1/\epsilon_2$. The Virasoro algebra is a subalgebra of $\mathcal{W}_{1+\infty}$, hence of $G(\mathbb{C}^3)$. The AGT correspondence IS the statement that Z_{Nek} is a character of a vertex algebra (a component of $G(X)$).

4.5 The AGT correspondence and $\mathcal{W}$-algebras

The Alday–Gaiotto–Tachikawa correspondence provides the most direct link between Nekrasov partition functions and vertex algebras (components of quantum vertex chiral groups).

Theorem 4.9 (AGT correspondence). *For the 4d $\mathcal{N} = 2$ $\mathrm{SU}(N)$ gauge theory of class $\mathcal{S}[\mathrm{SU}(N), C_{g,n}]$ on a Riemann surface $C_{g,n}$ of genus g with n punctures:*

$$Z_{\text{Nek}}^{\mathrm{SU}(N)}(\epsilon_1, \epsilon_2, \vec{a}, \vec{m}; \mathfrak{q}) = \left\langle \prod_{i=1}^{n} V_{\alpha_i}(z_i) \right\rangle_{\mathcal{W}_N,\, c(\epsilon_1,\epsilon_2)}, \tag{32}$$

where the right side is a conformal block of the $\mathcal{W}_N$-algebra at central charge $c = (N-1)\left(1 + N(N+1)\left(\frac{\epsilon_1}{\epsilon_2} + \frac{\epsilon_2}{\epsilon_1} + 2\right)\right)$, vertex operators V_{α_i} are determined by the puncture data, and $\vec{m}$ are the mass parameters.

In the Volume III framework, the AGT correspondence is the statement that the Nekrasov partition function is a genus-g n-point function of the quantum chiral algebra A_X ($\mathcal{W}_N$ sector of $G(X)$). The genus expansion of Z_{Nek} is the shadow obstruction tower:

- Genus 0, no punctures: the prepotential $\mathcal{F}_0$ = arity-2 shadow (κ);
- Genus 0, n punctures: the n-point tree-level amplitude = arity-n shadow;
- Genus g: the genus-g free energy $\mathcal{F}_g = \mathrm{obs}_g = \kappa \cdot \lambda_g$.

5 Synthesis: the four-layer dictionary

We now assemble the complete dictionary linking the four layers.

5.1 The master dictionary

4d $\mathcal{N}=2$ physics	**Hitchin system**	**KS wall-crossing**	**Quantum vertex chiral group** $G(X)$
CY3 X	Curve C, group G	CY3 category $\mathcal{C}_X$	Root datum $\mathcal{R}(X)$
Coulomb branch $\mathcal{M}_{\text{Coulomb}}$	Hitchin base $\mathcal{A}_H$	Stability conditions $\mathrm{Stab}(\mathcal{C}_X)$	Complex structure moduli
BPS state of charge γ	Point on Hitchin fiber	Stable object of class γ	Root vector e_γ
BPS index $\Omega(\gamma; u)$	DT invariant	Numerical DT	Root multiplicity $\mathrm{mult}(\gamma)$
Central charge $Z_\gamma(u)$	Period of λ_{SW}	Stability function	Central charge of Θ_{A_X}
Wall of marginal stability	Discriminant locus	Wall in $\mathrm{Stab}(\mathcal{C}_X)$	Gauge transformation locus
Wall-crossing formula	Monodromy of periods	KS product invariance	Gauge equiv. of MC elements
SW curve Σ	Spectral curve	Heart of t-structure	Classical limit of $G(X)$
SW differential λ_{SW}	Tautological 1-form	Central charge map	Tree-level R-matrix data
Prepotential $\mathcal{F}(\vec{a})$	Abel–Jacobi map	Numerical DT generating fn	Arity-2 shadow (κ)
$Z_{\text{Nek}}(\epsilon_1, \epsilon_2, \vec{a})$	Refined counting	Motivic DT	Character of $G(X)$
NS limit $\epsilon_2 \to 0$	Quantization of spectral curve	Quantum DT	Quantum curve = oper
AGT conformal block	Hitchin–$\mathcal{W}_N$	Categorical Hall algebra	n-point function of A_X

5.2 The structural pipeline

For a CY3 X (compact or local), the complete pipeline is:

$$\boxed{\begin{aligned} &X \ (\text{CY3}) \\ &\quad \xrightarrow{\text{categorical}} \mathcal{C}_X = D^b(\mathrm{Coh}(X)) \text{ or } \mathrm{Fuk}(X) \\ &\quad \xrightarrow{\Phi} A_X = \Phi(\mathcal{C}_X) \text{ (quantum chiral algebra)} \\ &\quad \xrightarrow{\text{root datum}} G(X) \text{ (quantum vertex chiral group)} \\ &\quad \xrightarrow{\text{character}} Z_{\text{Nek}}(\epsilon_1, \epsilon_2, \vec{a}) = \chi_{G(X)}(\mathcal{F}_{\vec{a}}) \end{aligned}} \tag{33}$$

Each arrow is a theorem or conjecture of this programme:

(1) $X \to \mathcal{C}_X$: the derived/Fukaya category of a CY3 is a CY category of dimension 3 (standard, proved by Kontsevich, Costello, and others);

(2) $\mathcal{C}_X \to A_X$: the CY-to-chiral functor Φ (Theorem CY-A of Volume III);

(3) $A_X \to G(X)$: extraction of the root datum $\mathcal{R}(X)$ from the quantum chiral algebra, identifying the BKM/Yangian structure (Theorem CY-C);

(4) $G(X) \to Z_{\text{Nek}}$: the character map (Conjecture **??**, proved for toric CY3 via AGT + Schiffmann–Vasserot).

5.3 Geometric Langlands and opers

The NS limit $\epsilon_2 \to 0$ connects to the geometric Langlands programme. The quantum spectral curve (**??**) is a ${}^L G$-oper on C: a flat ${}^L G$-connection with a specific reduction to a Borel subgroup.

- The Hitchin fibration $\pi\colon \mathcal{M}_H(C, G) \to \mathcal{A}_H$ on the A-side (Higgs bundles) corresponds to the space of opers on the B-side (flat connections with the Langlands dual group ${}^L G$).
- The quantum curve at $\hbar = \epsilon_1$ is an oper with "quantum parameter" $\hbar$. At $\hbar = 0$ one recovers the classical spectral curve; at $\hbar \neq 0$ one obtains a D-module on C.
- The Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}})$ at the critical level $k = -h^\vee$ parametrizes ${}^L G$-opers. The NS partition function is a section of the line bundle on the space of opers determined by the $\mathcal{W}$-algebra.
- In the Volume III framework: the passage from the spectral curve (classical) to the oper (quantum) is the passage from the tree-level R-matrix $r(z)$ to the full quantum R-matrix $R(z, \hbar)$ of $G(X)$. The parameter $\hbar = \epsilon_1$ controls the "quantum depth" in the E_2-chiral hierarchy.

Conjecture 5.1 (Quantum vertex chiral Langlands)**.** *The geometric Langlands correspondence for a curve C and gauge group G, in the "quantum" ($\hbar \neq 0$) regime, is the equivalence of representation categories of quantum vertex chiral groups:*

$$\operatorname{Rep}^{E_2}(G(X_{G,C})) \simeq D\text{-mod}(\operatorname{Bun}_{{}^L G}(C)) \tag{34}$$

where $X_{G,C}$ is the local CY3 engineering the class $\mathcal{S}$ theory, and D-mod$(\operatorname{Bun}_{{}^L G}(C))$ is the category of D-modules on the moduli of ${}^L G$-bundles on C (with the quantum parameter $\hbar$ playing the role of the "level" of the D-module category).

Remark 5.2. The three limits of (ϵ_1, ϵ_2) correspond to three regimes of the Langlands programme:

- $\epsilon_1 = \epsilon_2 = 0$: classical geometric Langlands (Beilinson–Drinfeld);
- $\epsilon_2 = 0$, $\epsilon_1 = \hbar \neq 0$: quantum geometric Langlands (the NS limit, opers, $\mathcal{W}$-algebra representations);
- ϵ_1, ϵ_2 both nonzero: the "doubly quantum" regime (the full Ω-background, motivic DT, K-theoretic Langlands of Aganagic–Frenkel–Okounkov).

Volume III operates primarily in the second and third regimes; the first is the "classical limit" where the quantum vertex chiral group degenerates to the classical one.

6 Summary and open problems

6.1 What has been established

(1) Type IIB on a CY3 X produces a 4d $\mathcal{N}=2$ theory whose Coulomb branch is $\mathcal{M}_{\mathrm{cx}}(X)$ and whose BPS spectrum comes from D3-branes on special Lagrangians (Section **??**).

(2) For local CY3 geometries (class $\mathcal{S}$ theories), the SW curve is the spectral curve of the Hitchin system, and the Hitchin fibration parametrizes the family of SW curves over the Coulomb branch (Section **??**).

(3) The KS wall-crossing formula governs BPS spectra across walls. In our framework this is gauge equivalence of MC elements Θ_u in the scattering diagram L_∞-algebra, and the KS product $\mathcal{A}(u)$ is the gauge-invariant shadow obstruction tower (Section **??**).

(4) The Nekrasov partition function is a refinement of the CY3 partition function. The NS limit gives the quantum curve (quantization of the spectral curve). Z_{Nek} should be the character of the quantum vertex chiral group $G(X)$ (Section **??**).

6.2 Open problems

OP1. Non-toric CY3. For CY3s that are neither toric nor of the form $K3 \times E$ (e.g. the quintic, complete intersections in toric varieties), the quantum vertex chiral group $G(X)$ is not yet identified. What is the “root datum” of the quintic? Is there still a BKM/Yangian structure?

OP2. Compact CY3. The present framework is most developed for local (noncompact) CY3 geometries. For compact CY3, the graviphoton and the universal hypermultiplet introduce additional structure. How does $G(X)$ incorporate gravitational corrections?

OP3. Wall-crossing and the full MC element. The identification of KS wall-crossing with gauge equivalence of Θ_u (Proposition **??**) needs a rigorous proof at the chain level, beyond the motivic/numerical shadow. This requires the full L_∞-structure of the scattering diagram algebra.

OP4. The refined topological vertex and motivic DT. The refined topological vertex computes Z_{Nek} combinatorially for toric CY3. The motivic DT invariants of KS should provide the “motivic character” of $G(X)$. Making this precise requires the motivic Hall algebra to be identified with the chain-level quantum vertex chiral group.

OP5. Geometric Langlands at the quantum level. Conjecture **??** connects $G(X)$ to the quantum geometric Langlands programme. The ϵ_1-dependence of the NS partition function should match the “level” of the quantum D-module category. The full two-parameter case ($\epsilon_1, \epsilon_2 \neq 0$) should correspond to the K-theoretic Langlands correspondence.

OP6. Physical unitarity and the BPS Hilbert space. The Fock space $\mathcal{F}_{\vec{a}}$ on which $G(X)$ acts should carry a positive-definite inner product (from the physical Hilbert space). This imposes constraints on $G(X)$ (unitarity bounds, signature conditions on the root lattice). What are the unitarity conditions on quantum vertex chiral groups?